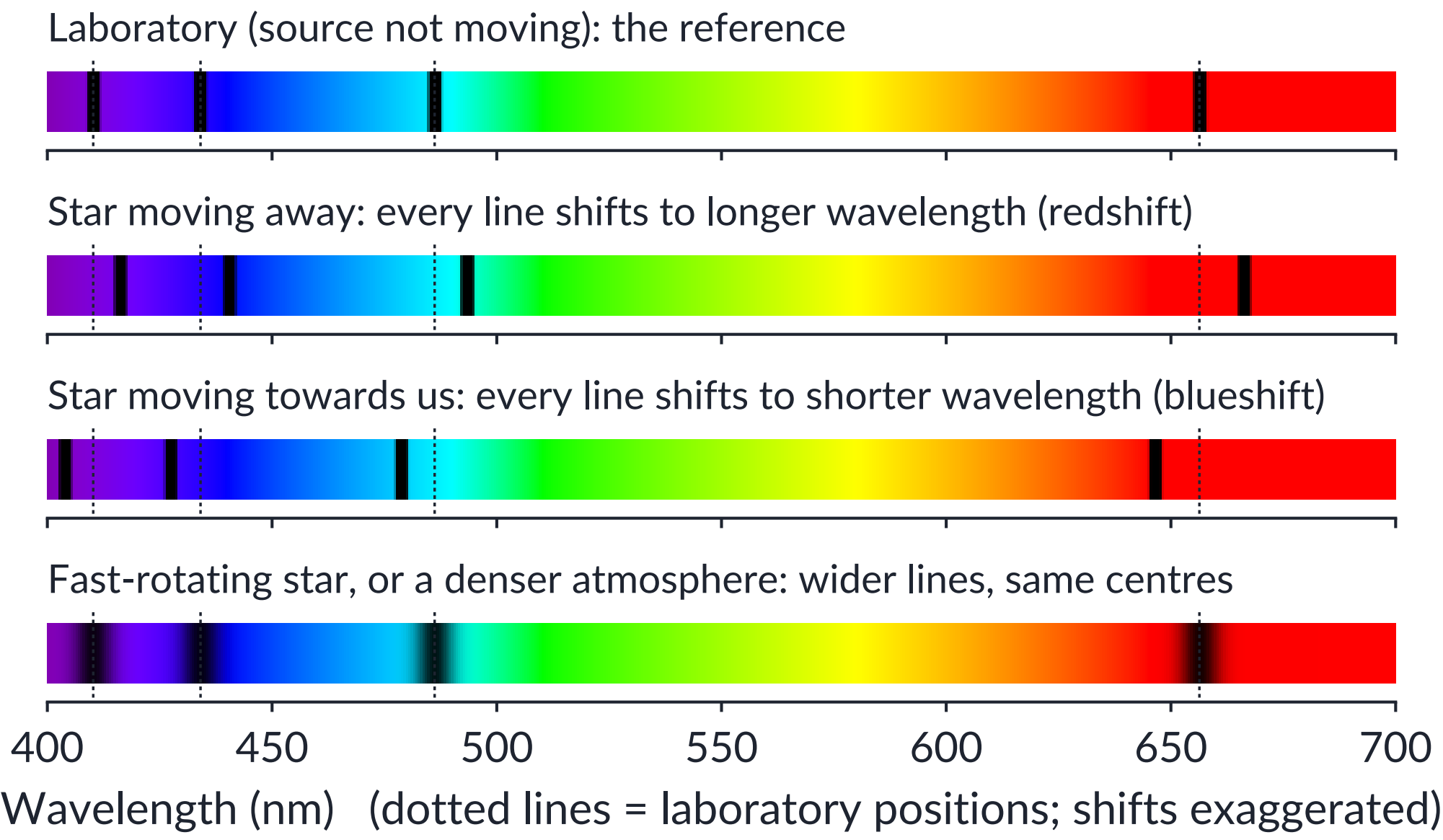
Laboratory (source not moving): the reference
Star moving away: every line shifts to longer wavelength (redshift)
Star moving towards us: every line shifts to shorter wavelength (blueshift)
Fast-rotating star, or a denser atmosphere: wider lines, same centres
400
450
500
550
600
650
700
Wavelength (nm) (dotted lines = laboratory positions; shifts exaggerated)